Department of Artificial Intelligence and Mathematical Sciences

Lab Manual

Natural Language processing (Course code)

Student name (Student ID)

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This is to certify that Mr. / Ms. ________________________ SO / DO _______________________ having Roll No. _______________ has successfully completed laboratory work during Spring Semester 2026.

Teacher Sign:

11	• **Lab # 11: Word Embeddings** • Introduction to Word Embeddings • Word2Vec Concept • Using Pretrained Embeddings	
12	**Lab # 12: Mini NLP Project** • Text Cleaning • Feature Extraction • Model Training • Model Testing and Result Analysis	

Lab Experiment # 01

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing by Removing Punctuation

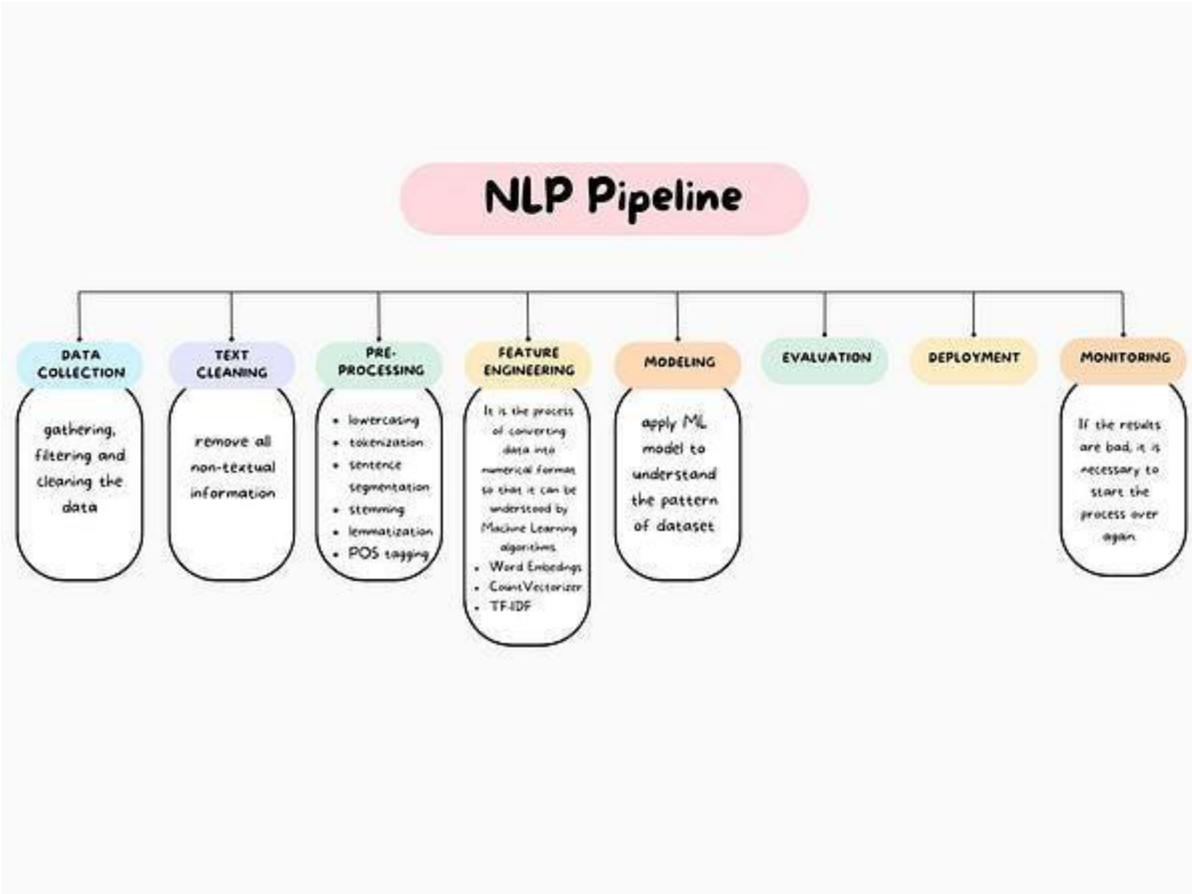
Objective:
• To understand the concept of **Natural Language Processing (NLP)** and its real-life applications such as voice assistants, chatbots, and text translation.
• To learn the importance of **text preprocessing in NLP**, especially removing punctuation to clean data and improve text analysis accuracy.

What is NLP ?

Natural Language Processing (NLP) is a branch of AI that enables computers to read, understand, and interpret human language.

Real life examples of NLP .

Voice assistants , chatbot , translation of text though software's



Removing Punctuation in NLP

Removing punctuation means deleting symbols like:
. , ! ? : ; ' " ( ) from text before processing it.
Example:
Original sentence:
"Hello, how are you?"
After removing punctuation:
Hello how are you

 Cleaner Text Data
Removing punctuation makes the text **clean and easier to process**.

 Reduces Noise in Data
Symbols like ! , . ? do not usually add meaning, so removing them **reduces unnecessary information**.

 Better Word Matching
Words like **“hello”** and **“hello!”** become the same after removing punctuation.

 Improves Model Accuracy
The model focuses on **important words instead of symbols**, which can improve results.

 Easier Tokenization
Splitting sentences into words becomes **simpler and more accurate**.

 Reduces Vocabulary Size
Without punctuation, the number of **different tokens (words)** becomes smaller.

Tokenization :

Tokenization is the process of breaking down the text into smaller units called tokens, usually words or phrases.

***Example**: The sentence “I love NLP” would be tokenized into [“I”, “love”, “NLP”]. Each word becomes a separate token.*

Lab task :

1:Install python jupyter notebook and install the nltk pandas
2 : take a paragraph of 100 words or 200 words and words then remove the punctuation from the paragraph
2: Perform tokenization of the paragraph taken

Lab Experiment # 02

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing Techniques

Objective:
• To understand the basic concept of Natural Language Processing (NLP) and its real-life applications such as voice assistants, chatbots, and translation systems.
• To learn different text preprocessing techniques like lowercasing, removing punctuation, special characters, and stop words to clean and prepare text data for analysis.

Stop Words Removal

Stop words are very common words that usually do not add important meaning to a sentence.
Examples include: **is, the, a, an, in, on, and, of**

These words are removed to make the text smaller and easier to analyze.

Example

Original sentence:
`This is a book on Natural Language Processing`

After removing stop words:
`book Natural Language Processing`

Lowercasing

Lowercasing means converting all letters in the text to **small letters**.

This helps the computer treat words like **Hello, HELLO, hello** as the same word.

Example

Original text:
`Hello World`

After lowercasing:
`hello world`

Removing Special Characters

Special characters are symbols that usually do not add meaning in text analysis.

Examples include:
`@ # $ % ^ & * ( ) _ +`

Removing them makes the text cleaner.

Example

Original sentence:
`Welcome!!! to NLP @ 2025`

After removing special characters:
`Welcome to NLP 2025`

Text Normalization

Text normalization means converting text into a **standard and consistent form** so that it becomes easier for the computer to understand.

It may include:

- Lowercasing
- Removing punctuation
- Removing special characters
- Expanding short forms

Example

Original text:

```
I'm learning NLP!!!
```

Normalized text:

```
i am learning nlp
```

Lab Task

Write a Python program to perform the following **text preprocessing steps**:

1. Take a sentence as input.
2. Convert the text into **lowercase**.
3. Remove **punctuation and special characters**.
4. Remove **stop words** from the sentence.
5. Display the **cleaned text** after preprocessing.

Example Input:

```
Hello! I am learning Natural Language Processing.
```

Expected Output:

```
learning natural language processing
```

Lab Experiment # 03

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing Techniques

Objective:
• To understand the basic concept of Natural Language Processing (NLP) and its real-life applications

Stemming

Stemming is a text preprocessing technique that **reduces words to their root or base form** by chopping off suffixes.
It may not produce a real word but helps in grouping similar words together.

Example:

- `running` → `run`
- `studies` → `studi`
- `happily` → `happi`

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

words = ["running", "studying", "smiling", "communication"]

stemmed_words = [stemmer.stem(word) for word in words]

print(stemmed_words)
```

outputs : *['run', 'studi', 'smile', 'commun']*

Lemmatization(grammar)

Lemmatization also reduces words to their root form, but it **produces a valid word (lemma)** using dictionary knowledge and context.

Example:

- `running` → `run`
- `studies` → `study`
- `better` → `good`

```
from nltk.stem import WordNetLemmatizer

from nltk.corpus import wordnet

lemmatizer = WordNetLemmatizer()

words = ["running", "better", "studies", "was"]

lemmas = [

    lemmatizer.lemmatize("running", pos=wordnet.VERB),

    lemmatizer.lemmatize("better", pos=wordnet.ADJ),

    lemmatizer.lemmatize("studies", pos=wordnet.NOUN),

    lemmatizer.lemmatize("was", pos=wordnet.VERB)

]

print(lemmas)
```

Lab task :

- Remove punctuation from the paragraph (200 words)
- Covert into lower
- Turn the paragraph into tokens
- Perform stemming and lemmatization on a paragraph or 200 words

Lab Experiment # 04

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing Techniques

Objective:
To understand and implement Part-of-Speech (POS) Tagging in Python to identify grammatical categories of words in a sentence.

Post tagging

POS Tagging (Part-of-Speech Tagging) is a process in Natural Language Processing (NLP) where each word in a sentence is assigned a grammatical category.

These categories help the computer understand the structure and meaning of a sentence**.**

Tag	Meaning	Example
NN	Noun	Ali, Car
VB	Verb	Run, Eat
JJ	Adjective	Beautiful
RB	Adverb	Quickly
PRP	Pronoun	He, She
DT	Determiner	The, A

Tag	Meaning	Example
NN	Noun	Ali, Car
VB	Verb	Run, Eat
JJ	Adjective	Beautiful
RB	Adverb	Quickly
PRP	Pronoun	He, She
DT	Determiner	The, A

Code ;

Required Libraries

```
import nltk
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')



import nltk
from nltk.tokenize import word_tokenize
from nltk import pos_tag

sentence = "Ali is playing football in the park"

# Tokenization
words = word_tokenize(sentence)

# POS Tagging
pos_tags = pos_tag(words)

print(pos_tags)
```

CODE FOR POS Tagging

```
import nltk
from nltk.tokenize import word_tokenize
from nltk import pos_tag

sentence = "Ali is playing football in the park"

# Tokenization
words = word_tokenize(sentence)

# POS Tagging
pos_tags = pos_tag(words)

print(pos_tags)
```

OUTPUT

```
[('Ali', 'NNP'),
 ('is', 'VBZ'),
 ('playing', 'VBG'),
 ('football', 'NN'),
 ('in', 'IN'),
```

('the', 'DT'),
('park', 'NN')]

Lab 4 task :

1. Write a Python program to perform POS tagging .

- "Sara is reading a book." Write your own sentences

2. Take a sentence input from the user and display POS tags for each word.
3. Write a program that counts the number of nouns and verbs in a sentence. (counter function will be used) from collections import Counter
4. Perform POS tagging on the sentence:

- "The quick brown fox jumps over the lazy dog."
- Identify:
- Nouns
- Verbs
- Adjectives

]

Lab Experiment # 05

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing Techniques

Objective:
To identify and classify named entities such as **Person, Location, and Organization** from text using Python.

Introduction

Named Entity Recognition (NER) is a task in **Natural Language Processing (NLP)** that identifies important entities in a sentence and classifies them into categories such as:

- **Person**
- **Location**
- **Organization**
- **Date**
- **Time**

NER helps computers understand **who, where, and what** is mentioned in the text.

NER is used in many applications such as:

- Search engines
- Chatbots
- News analysis
- Information extraction

"Elon Musk is the CEO of Tesla and he lives in the United States."

NER Output:

Word	Entity Type
Elon Musk	Person
Tesla	Organization
United States	Location

Installation libraries

- `pip install spacy` → installs the spaCy library used for Named Entity Recognition.
- `pip install nltk` → installs the NLTK library for other NLP tasks like tokenization and POS tagging.
- `pip install notebook` → installs Jupyter Notebook to run Python code interactively.
- `python -m spacy download en_core_web_sm` → downloads the English language model required for NER.

Required Libraries (Installation Steps)

```
import spacy              # Import the spaCy library used for NLP tasks

# Load the English language model
nlp = spacy.load("en_core_web_sm")

# Define a sentence that we want to analyze
text = "Elon Musk founded Tesla in the United States."

# Process the text using the spaCy NLP pipeline
doc = nlp(text)

# Loop through all detected named entities in the sentence
for entity in doc.ents:

    # Print the entity text and its entity label
    print(entity.text, entity.label_)
```

OUTPUT

Elon Musk PERSON
Tesla ORG
United States GPE

Function / Method	Description	Example Use
`import spacy`	Imports the spaCy library for NLP tasks	`import spacy`
`spacy.load()`	Loads a trained language model	`nlp = spacy.load("en_core_web_sm")`
`nlp()`	Processes the text using the NLP pipeline	`doc = nlp(text)`
`doc.ents`	Returns all named entities detected in the text	`for ent in doc.ents:`
`ent.text`	Displays the actual entity word or phrase	`print(ent.text)`
`ent.label_`	Displays the category of the entity (PERSON, ORG, etc.)	`print(ent.label_)`
`spacy.displacy.render()`	Visualizes named entities in the text	`displacy.render(doc, style="ent")`

Lab tasks

Task 1: Perform NER on a Simple Sentence

Sentence: "Bill Gates founded Microsoft in the United States."

Steps:

1. Import the **spaCy** library.
2. Load the **English language model** (`en_core_web_sm`).
3. Store the sentence in a variable.
4. Process the sentence using the **nlp()** function.
5. Print all detected **named entities and their labels**.

Task 2: NER on User Input

Steps:

1. Import **spaCy**.
2. Load the English model.
3. Ask the user to **enter a sentence** using `input()`.

4. Process the text using **nlp()**.
5. Print each **entity and its type**.

Example:

```
Enter a sentence: Elon Musk founded SpaceX in California.
```

Task 3: Identify Specific Entity Types

Sentence: "Imran Khan was born in Lahore and served as Prime Minister of Pakistan."

Steps:

1. Import spaCy.
2. Load the English model.
3. Store the sentence in a variable.
4. Extract entities using **doc.ents**.
5. Print entities that belong to:
   - **PERSON**
   - **LOCATION / GPE**
   - **ORGANIZATION / ORG**

Task 4: Count Total Named Entities

Sentence: "Apple released the iPhone in California."

Steps:

1. Import spaCy.
2. Load the English model.
3. Store the sentence in a variable.
4. Process the text using **nlp()**.
5. Use `len(doc.ents)` to count total entities.
6. Print the total number of entities.

Example Output:

```
Total Entities Found: 2
```

Task 5: NER on a Paragraph

Paragraph: "Elon Musk is the CEO of Tesla. He lives in the United States and often visits California."

Steps:

1. Import the spaCy library.
2. Load the English language model.
3. Store the paragraph in a variable.
4. Process the paragraph using **nlp()**.
5. Loop through **doc.ents**.
6. Print all detected entities with their labels.

Lab Experiment # 06

Title: Introduction to Natural Language Processing (NLP) and Text Preprocessing Techniques

Objective:
To identify and classify named entities such as **Person, Location, and Organization** from text using Python.

Bag of words (BoW) model in NLP